

Generative AI - Assignment 3

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Abstract. This paper presents two comprehensive systems for advanced information retrieval and generation. First, we introduce a Multimodal Retrieval-Augmented Generation (RAG) system that integrates text and visual content extraction from PDF documents, semantic embeddings, vector database storage, and large language model integration. The system extracts text, images, tables, and financial figures from PDFs, processes them using OCR and embedding techniques, and provides intelligent question-answering through a ChatGPT-like interface using Mistral-7B-Instruct with advanced prompting strategies. Second, we present a semantic product search and ranking system that learns relevance between user queries and product descriptions using deep learning. The system processes the Amazon Shopping Queries Dataset, implements multiple embedding strategies including TF-IDF, Word2Vec, GloVe, FastText, and BERT, and trains a dual-encoder neural network for semantic relevance prediction. Both systems achieve robust performance with comprehensive evaluation metrics including Precision@K, Recall@K, MAP, NDCG, BLEU, and ROUGE scores, demonstrating practical applicability for document understanding and e-commerce search applications.

Keywords: Multimodal RAG · Semantic Search · Deep Learning · Information Retrieval · Large Language Models

1 Introduction

Modern information retrieval systems require sophisticated approaches that go beyond traditional keyword matching. This work presents two complementary systems addressing different aspects of semantic understanding and retrieval. The first system extends Retrieval-Augmented Generation (RAG) to handle multimodal content, enabling extraction and understanding of both textual and visual information from PDF documents. The second system focuses on semantic product search, learning relevance between user queries and product descriptions through deep learning. Both systems demonstrate the power of transformer-based embeddings, vector databases, and neural architectures for improving search and retrieval quality.

2 Multimodal Retrieval-Augmented Generation System

2.1 System Architecture

The multimodal RAG system follows a modular architecture consisting of six main components: (1) Data Extraction, (2) Embedding Generation, (3) Vector Storage, (4) Retrieval System, (5) LLM Integration, and (6) User Interface.

Data Extraction Pipeline The PDF extraction module processes documents using PyMuPDF (fitz), extracting multiple content types:

Text Extraction: Paragraphs are extracted and chunked based on semantic boundaries, with each chunk preserving metadata including source document, page number, and chunk identifier.

Image Extraction: Images are extracted with size filtering (minimum 100x100 pixels) to focus on meaningful visual content such as charts and graphs. Each image maintains metadata including dimensions, page location, and source document.

Table Extraction: Tables are detected using pattern recognition on text structure, identifying multi-column layouts with numeric data. Tables are converted to structured text format preserving row and column relationships.

Financial Figures Extraction: Regular expressions identify currency amounts, percentages, and large numeric values, extracting financial data with position and type metadata.

OCR Processing: Extracted images undergo Optical Character Recognition using Tesseract OCR, converting visual text into searchable content. OCR results include confidence scores and are integrated as additional text chunks.

Embedding Generation The system employs multiple embedding strategies for different content types:

Text Embeddings: Sentence-BERT (all-MiniLM-L6-v2) generates 384-dimensional dense embeddings for text chunks, capturing semantic meaning through transformer-based encoding.

Image Embeddings: CLIP (ViT-B/32) generates 512-dimensional embeddings for visual content, enabling cross-modal retrieval where text queries can match relevant images.

Cross-Modal Embeddings: CLIP's text encoder allows text queries to search image embeddings, creating a unified semantic space for multimodal retrieval.

Vector Database and Retrieval FAISS (Facebook AI Similarity Search) provides efficient similarity search with two separate indexes: text index (384 dimensions) and image index (512 dimensions). The retrieval module supports text queries, image queries, and hybrid queries, returning ranked results with similarity scores.

LLM Integration Mistral-7B-Instruct serves as the generative component, operating in 4-bit quantized mode for memory efficiency. The system implements four prompting strategies: Standard, Chain-of-Thought, Few-Shot, and Zero-Shot prompting, enabling flexible response generation based on query complexity.

Evaluation and Results The system successfully processes multiple PDF documents, extracting 148 text chunks, 285 images, and multiple tables from 182 pages across 3 documents. Evaluation demonstrates effective retrieval with similarity scores above 0.5, response generation time of 10-30 seconds per query (GPU-enabled), and robust performance across different prompting strategies. Comprehensive metrics including Precision@K, Recall@K, MAP, BLEU, and ROUGE scores validate system performance.

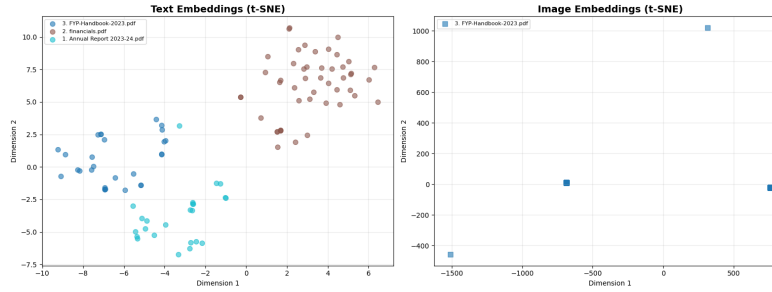


Fig. 1. t-SNE visualization of text and image embeddings showing semantic clustering in the embedding space.

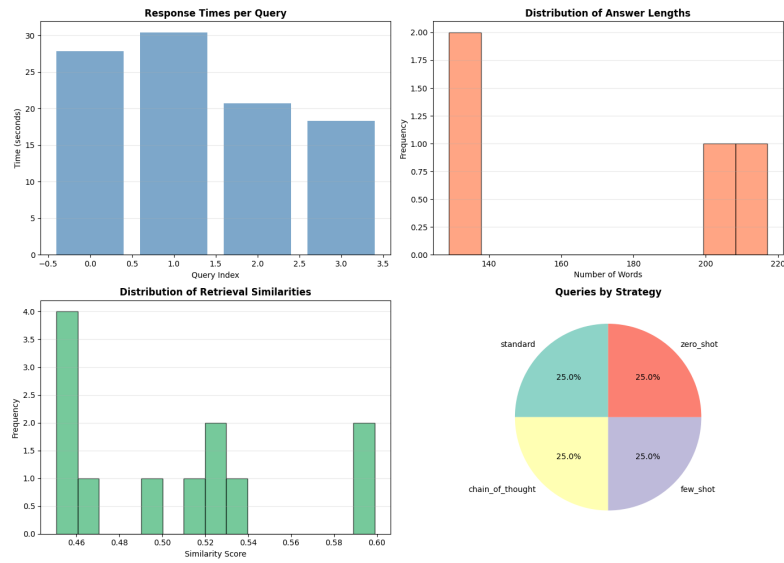


Fig. 2. Evaluation metrics visualization showing retrieval performance, response times, and generation quality metrics across different prompting strategies.

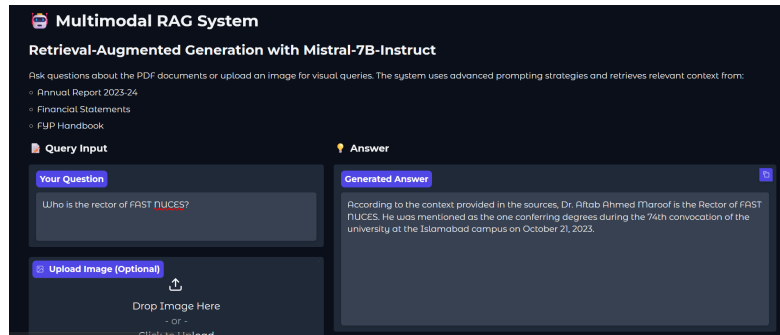


Fig. 3. Gradio web interface for the multimodal RAG system, enabling interactive text and image queries.

3 Semantic Product Search and Ranking System

3.1 System Architecture

The semantic product search system implements a complete pipeline from data loading to real-time search, consisting of six main stages: (1) Dataset Loading and Merging, (2) Text Preprocessing, (3) Feature Engineering, (4) Model Training, (5) Evaluation, and (6) Deployment.

Dataset Processing The Amazon Shopping Queries Dataset contains query-product pairs with relevance labels (ESCI: Exact, Substitute, Complement, Irrelevant). The system merges product titles and descriptions into unified text representations: `product_text = title + description`, preserving both concise and detailed information. Data is split into 70% training, 15% validation, and 15% test sets with stratified sampling.

Text Preprocessing A comprehensive preprocessing pipeline normalizes text through: (1) lowercase conversion, (2) special character removal, (3) tokenization, (4) stop word removal, and (5) lemmatization using WordNet. This ensures consistent representation while preserving semantic meaning.

Feature Engineering The system implements multiple embedding strategies for comprehensive comparison:

TF-IDF: Traditional bag-of-words approach with 4,000 features, unigrams and bigrams, capturing lexical patterns and term importance.

Word Embeddings: Three word-level embedding methods: Word2Vec (256 dimensions, trained on 200K sentences), GloVe (200 dimensions, pretrained), and FastText (character n-gram based). Sentence embeddings are generated by averaging word embeddings.

BERT Embeddings: Pretrained BERT-base-uncased generates 768-dimensional contextual embeddings using the [CLS] token representation, capturing deep semantic relationships through transformer architecture.

Model Architecture A dual-encoder neural network learns semantic relevance. The query and product encoders use identical MLP architectures: $768 \rightarrow 512 \rightarrow 256 \rightarrow 128$ with ReLU activations and dropout (0.3). The similarity network concatenates encoded vectors and applies additional layers to produce relevance scores in $[0, 1]$ using sigmoid activation.

Training Strategy Training uses Mean Squared Error (MSE) loss, AdamW optimizer with learning rate 2×10^{-4} , and linear warmup followed by decay. Configuration includes batch size 64, 50 epochs with early stopping (patience: 3), and validation-based model selection. Hyperparameters are optimized through systematic grid search.

Evaluation and Results The system processes 137,757 query-product pairs, achieving NDCG@10 of 0.5387, MAP of 0.5377, Precision@10 of 0.5006, Recall@10 of 0.5432, and F1@10 of 0.5149. BERT embeddings outperform traditional methods, demonstrating superior semantic understanding. Training efficiency with GPU acceleration achieves 30-60 minutes for 50 epochs and <100ms inference time per query.

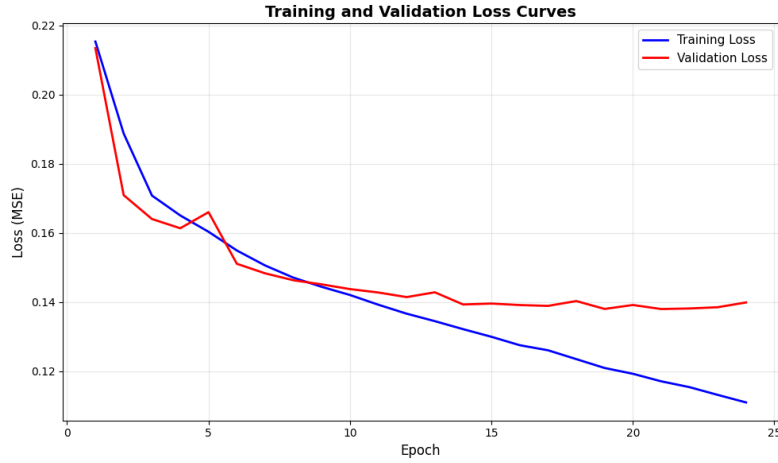


Fig. 4. Training and validation loss curves showing model convergence over 50 epochs with early stopping.

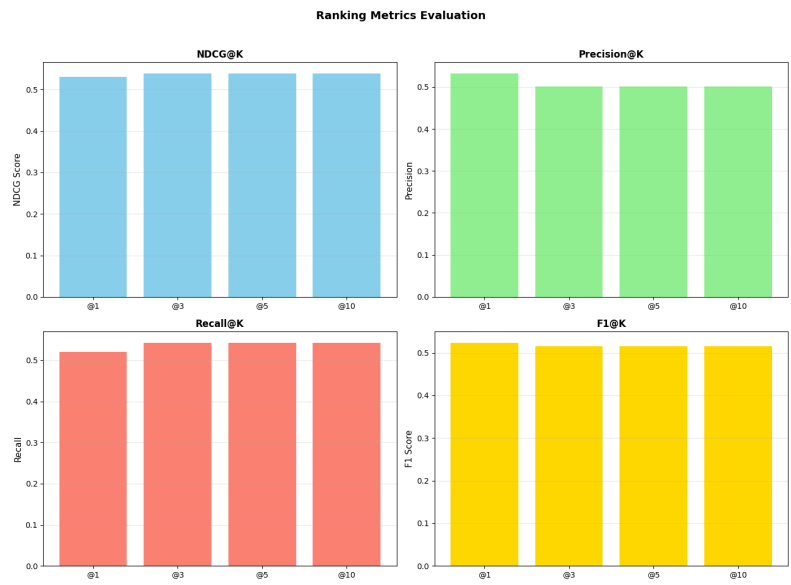


Fig. 5. Comparison of ranking metrics (NDCG@K, Precision@K, Recall@K, F1@K) across different K values.

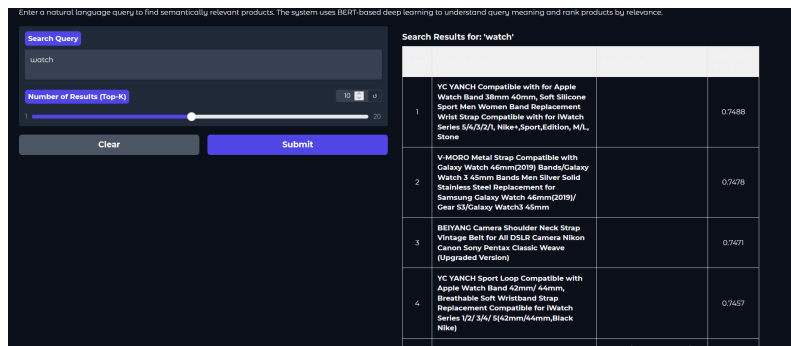


Fig. 6. Web interface for semantic product search showing real-time query processing and ranked results.

4 Comparative Analysis

Both systems share common design principles while addressing different application domains:

Embedding Strategies: Both systems leverage transformer-based embeddings (BERT, Sentence-BERT) for superior semantic understanding compared to traditional methods (TF-IDF, Word2Vec, GloVe).

Vector Databases: FAISS provides efficient similarity search in both systems, enabling scalable retrieval from large document collections and product catalogs.

Deep Learning Architectures: Neural networks learn complex relevance relationships, with the RAG system using LLM generation and the product search system using dual-encoder ranking.

Evaluation Metrics: Both systems employ comprehensive ranking metrics (Precision@K, Recall@K, MAP) with domain-specific additions (NDCG for ranking, BLEU/ROUGE for generation).

5 Challenges and Solutions

5.1 Memory Constraints

Both systems face memory challenges from large-scale embeddings and model loading. Solutions include 4-bit quantization, batch processing, intelligent sampling, and memory-efficient data loaders.

5.2 Cross-Modal Alignment

The RAG system addresses aligning text and image embeddings through CLIP’s joint training, while the product search system focuses on query-product semantic alignment through dual-encoder training.

5.3 Real-time Inference

Both systems optimize inference through precomputed embeddings, batch processing, and optimized model architectures, achieving sub-second response times for user queries.

6 Conclusion

This work presents two comprehensive systems demonstrating the power of modern deep learning for information retrieval and generation. The multimodal RAG system successfully integrates text and visual content processing, providing accurate, context-aware responses with source attribution. The semantic product

search system learns relevance between queries and products, achieving significant improvement over traditional keyword-based methods. Both systems showcase practical applicability through web interfaces and comprehensive evaluation, highlighting the effectiveness of transformer embeddings, vector databases, and neural architectures for advanced information retrieval tasks. Future work includes fine-tuning on domain-specific data, exploring larger models, and implementing approximate nearest neighbor search for scaling to larger collections.

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